INSTITUTE OF ENGINEERING & MANAGEMENT, KOLKATA
(School of University of Engineering and Management, Kolkata)

17th Nov. 25

End Semester Examination, November - December, 2025

Degree: **B.Tech** Semester: **First**

Stream: **BT / CSE(AI)/ CSE(AI & ML)/ CSE(Cybersecurity)/ CSE (IOT, CS, BT)/ CSE(Networks)/ CSBS/ CSE/ ECE/ EEE/ EE/ IT/ ME**

Paper Code: **BSCCH102**

Paper Name: **Chemistry**

Full Marks: **100** Duration: **3 Hours**

Part - A
Attempt any 10 questions out of 15 questions
Each question carries 2 Marks (2 × 10)

1. What is the significance of Ψ^2 in quantum mechanical model of atom? 2

2. How many kinds of H^1 are in 2
(i) $CH_2 = CH_2$ (ii) $C_6H_5 - CH_3$ (iii) $CH_3 - CH = CH_2$?

3. Define molar conductance. Give its units. 2

4. Define Coordination number. 2

5. Mark the asymmetric carbon atoms and give the number of optical isomers in the compound $HOOC(CHOH)_2COOH$. 2

6. Is the breaking of C-H bond in nucleophilic substitution homolytic or heterolytic? 2

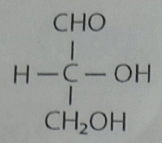
7. Why are Br_+ and CCl_2 electrophiles? 2

8. What is cell constant? Give its units. 2

9. Define Coordination sphere. 2

10. For UV radiation of 200 nm, what will be the wave number in cm^{-1}? 2

11. Assign R or S configuration of each of the following compounds: 2



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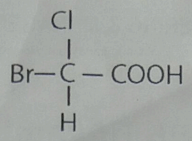

12. Is the following compound will exhibit optical isomerism? 2
$CH_3CH(OH)CH(OH)CH_3$

13. Which of the following compounds will show optical isomerism of 2
(i) H_2CH_2COOH (ii) $(CH_3)_2CHCHO$
(iii) $H_2NCH(CH_3)_2$ (iv) $CH_3CH(OH)COOH$

14. Discuss the term "carbanium ion". 2

15. Indicate clearly the reactants, transition state activation energies and products of the reaction. 2

Part - B
Attempt any 6 questions out of 9 questions
Each question carries 5 Marks (5 × 6)

16. Write down the Schrodinger wave equation and explain the significance of the term involved. 5

17. The moment of inertia of NO molecule is 16.5×10^{-4} gm. Cm2. What will be the energy of first rotational level (J=1)? 5

18. Define acids and bases in the light of Arrhenius theory. What are the limitations of this theory? 5

19. Write note on Optical isomerism. 5

20. Bond energies of two types of bonds are 242 kJ mol^{-1} and 1,150 kJ mol^{-1}. Which of these refers to the homolytic fission and which to the heterolytic fussion? 5

21. How will you synthesize: (i) salicyladhyde, (ii) salicylic acid from phenol? 5

22. For a gas T_C = 304.2 K, P_C =72.8 atm. Calculate vander Waals' constant for this unknown gas. 5

23. Derive Kirchoff's equation and show how the heat of reaction at constant volume is related at constant pressure. 5

24. What are intermolecular forces? Discuss the origin of intermolecular forces. 5

Part - C

Attempt any 5 questions out of 8 questions
Each question carries 10 Marks (10 × 5)

25. Argon has $T_c = -122^0C, P_c = 48\ atm$) What is the radius of the Argon atom? **10**

26. The boiling point of water at 50 atm pressure is 256°C. Compare the theoretical efficiencies of system of engine operating between 45°C and the boiling point of water at: (i) 1 atm, and (ii) 50 atm. **10**

27. (a) State and explain Carnot's theorem. **10**
(b) Deduce an expression for the efficiency of a Carnot's engine working between two temperatures T_1 *and* T_2.
(c) What should be the temperature of the sink for the efficiency to be unity?

28. What are first, second and third ionisation energies of an element? What are the factors on which ionisation energy of an element depends? **10**

29. Name the elements of symmetry. Discuss, with the help of an example, an optically active compound without chirality. **10**

30. What are the different types of organic reactions encountered in organic chemistry? Discuss in brief. **10**

31. Explain the bond order of N_2 molecule and CO molecule with the help of MO energy level diagram. How does molecular orbital differ from atomic orbital? On the basis of MO theory, account for the following: **10**
(i) Bond order of O_2^+ is more than O_2 molecule; whereas bond order of N_2^+ is less than N_2 molecule.
(ii) He_2 molecule does not exist.

32. What are the deactivation pathways of electronically excited molecules? Draw Jabloriski diagram and explain. **10**

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